Assignment 2 - Task 2

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State Diagram

$M = (Q, \Sigma, \delta, s, F)$

Q (States)

$Q = \{$StartTurn, RollingDice, HandlingRobber, CollectingResoures, ActionPhase, EndTurn, Invalid$\}$

Σ (Alphabet)

$\Sigma = \{$roll, dice=7, dice $\neq$ 7, robberHand, resourcesCollected, buildRoad, buildSettlement, buildCity, pass$\}$

s (Initial state)

$s = \{$StartTurn$\}$

F (Final States)

$F = \{$EndTurn$\}$

δ (Transition Function)

δ(StartTurn, roll) = RollingDice

δ(RollingDice, dice=7) = HandlingRobber

δ(RollingDice, dice $\neq$ 7) = Collecting Resources

δ(HandlingRobber, robberHandled) = CollectingResources

δ(CollectingResources, resourcesCollected) = Action Phase

δ(ActionPhase, buildRoad) = ActionPhase

δ(Action Phase, buildSettlement) = Action Phase

δ(ActionPhase, buildCity) = ActionPhase

δ(Action Phase, pass) = EndTurn